

LAWRENCE, KS, USA

WMO: 724508

Lat: 39.008N

Lon: 95.212W

Elev: 832

StdP: 14.26

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 03997

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.1	(c) 8.6	(d) -5.2	(e) 4.3	(f) 5.3	(g) -0.5	(h) 5.5	(i) 11.2	(j) 27.7	(k) 35.5	(l) 25.3	(m) 34.2	(n) 6.3	(o) 320	(p) 0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.5	98.8	76.7	95.1	76.2	91.8	75.5	79.9	92.2	78.4	90.6	77.2	88.9	10.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.4	142.0	86.8	75.0	135.2	85.5	73.4	128.3	83.6	44.0	92.4	42.6	91.1	41.2	88.8	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	21.8	19.1	-6.1	103.3	6.2	3.5	-10.6	105.7	-14.2	107.8	-17.7	109.7	-22.2	112.2
			-6.7	82.6	6.0	1.5	-11.0	83.7	-14.5	84.5	-17.9	85.4	-22.2	86.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.5	29.8	34.0	44.9	55.2	65.5	75.0	79.3	77.5	69.3	56.4	44.2	33.2
	DBStd	19.57	11.09	11.43	11.25	9.50	8.34	6.35	5.80	6.18	8.08	9.60	10.04	10.67
	HDD50	2173	627	459	232	54	3	0	0	0	0	45	226	527
	HDD65	4962	1090	867	628	315	101	6	0	1	46	298	625	985
	CDD50	4173	3	11	74	210	484	749	908	853	579	243	52	7
	CDD65	1486	0	0	5	21	116	305	443	389	175	31	1	0
	CDH74	16135	0	2	52	263	1145	3233	5086	4182	1841	318	13	0
	CDH80	7315	0	0	7	55	367	1383	2573	2078	764	87	1	0
Wind	WSAvg	8.4	8.8	9.1	10.1	10.7	9.0	8.2	6.4	6.2	7.0	8.1	8.8	8.6
Precipitation	PrecAvg	39.3	1.2	1.2	2.5	3.6	5.0	6.0	4.3	4.0	4.5	3.3	2.4	1.7
	PrecMax	59.3	3.5	3.1	10.6	8.9	11.5	16.3	18.3	10.9	12.8	10.1	6.2	4.8
	PrecMin	21.6	0.0	0.0	0.4	0.7	0.7	2.0	0.2	0.6	0.6	0.2	0.0	0.0
	PrecStd	9.6	0.9	0.8	1.8	2.2	2.5	3.7	3.4	2.6	3.2	2.2	1.7	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.4	71.5	80.0	86.1	91.2	98.3	102.4	103.2	97.2	87.8	75.4	66.4
		MCWB	50.3	56.1	62.7	66.3	72.6	76.2	75.5	75.5	73.8	70.6	61.7	57.6
	2%	DB	57.5	64.0	73.5	80.9	87.8	93.5	99.0	98.8	91.4	81.9	71.5	60.7
		MCWB	47.5	53.3	59.1	64.3	70.9	75.7	77.3	76.4	73.1	66.3	59.8	52.7
	5%	DB	52.4	57.9	69.7	76.7	84.1	90.8	95.4	94.5	88.3	76.9	65.9	54.8
		MCWB	44.0	48.6	57.1	62.3	69.4	75.1	77.4	75.8	73.3	64.1	55.6	47.8
	10%	DB	46.6	52.4	64.2	72.6	80.9	88.0	91.8	90.7	84.0	72.9	61.3	50.0
		MCWB	40.1	44.7	54.1	60.7	68.3	74.1	76.6	75.4	70.4	62.6	52.8	44.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.4	59.7	65.8	69.4	75.7	80.1	82.3	81.0	78.0	73.3	64.5	60.2
		MCDB	60.1	66.2	75.8	81.5	87.1	91.7	93.5	93.1	89.2	83.1	71.4	65.2
	2%	WB	48.8	54.2	62.6	67.1	73.4	78.1	80.3	79.2	75.7	70.5	60.5	54.1
		MCDB	56.0	61.7	70.4	76.5	83.7	89.9	92.6	92.3	87.4	78.2	68.1	58.4
	5%	WB	44.5	49.6	59.4	64.7	71.5	76.6	79.0	77.8	74.3	67.4	57.7	48.3
		MCDB	51.7	58.4	66.8	73.7	81.2	87.9	91.3	90.3	85.0	73.7	63.9	53.6
	10%	WB	40.4	44.7	54.8	62.0	69.4	75.0	77.8	76.3	72.5	64.1	54.0	44.1
		MCDB	47.2	52.1	63.2	71.3	78.3	85.5	89.5	87.7	82.1	70.5	60.6	49.9
Mean Daily Temperature Range	5% DB	MDBR	21.2	21.6	23.1	23.4	21.9	21.0	21.5	22.6	24.3	23.9	23.1	20.5
		MCDBR	30.7	31.5	30.9	29.2	25.3	23.3	25.4	27.1	26.1	27.8	29.2	28.6
		MCWBR	21.4	21.1	18.3	16.3	12.2	9.9	9.2	9.7	11.0	15.0	18.4	20.5
	5% WB	MCDBR	28.5	29.0	24.2	24.2	21.5	21.3	21.5	23.1	22.7	21.4	24.3	25.2
		MCWBR	20.6	20.6	15.7	15.3	11.8	10.5	9.8	10.3	10.6	13.7	17.4	19.9
Clear-Sky Solar Irradiance	taub	0.292	0.305	0.335	0.388	0.425	0.434	0.438	0.438	0.389	0.343	0.308	0.287	
	taud	2.474	2.447	2.422	2.292	2.246	2.251	2.266	2.272	2.358	2.475	2.515	2.510	
	Ebn at Noon	282	293	293	282	272	269	266	264	271	275	274	276	
	Edh at Noon	25	30	34	41	43	43	43	41	36	29	24	23	
All-Sky Solar Radiation	RadAvg	710	961	1245	1585	1819	2062	2049	1824	1521	1096	777	613	
	RadStd	71	106	104	103	160	145	134	94	104	124	72	57	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	-0.97	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page